

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Startegic Product Placement Analysis
Maximum Marks	4 Marks

Technical Architecture:

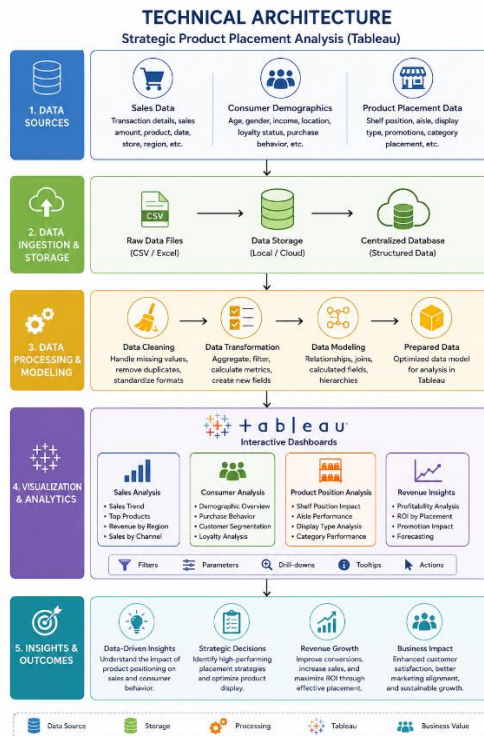


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards for users to view and analyze data	Tableau Dashboard
2	Application Logic-1	Data preprocessing and cleaning logic	Tableau
3	Application Logic-2	Data transformation and integration logic	Tableau Dashboard
4	Application Logic-3	Data analysis and visualization logic	Tableau
5	Database	Stores structured sales, product, and customer data	N/A
6	Cloud Database	Cloud-based storage for scalable data handling	N/A
7	File Storage	Stores datasets and exported reports	Local File System / Cloud Storage (AWS S3)
8	External API-1	Data import from external retail or analytics sources	Retail APIs / CSV Data Sources
9	External API-2	Optional integration for demographic or market data	N/A
10	Machine Learning Model	Predict trends and customer behavior patterns	N/A
11	Infrastructure (Server / Cloud)	Deployment and hosting environment	Local System / AWS / Azure

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Libraries used for data processing and analysis	Python (Pandas, NumPy, Matplotlib)
2	Security Implementations	Data protection and access control mechanisms	Authentication, Data Encryption
3	Scalable Architecture	Ability to handle increasing data volume	Cloud-based architecture (AWS, BigQuery)
4	Availability	Ensures system is accessible when needed	Cloud Hosting with High Availability
5	Performance	Efficient processing and fast dashboard loading	Tableau Optimization, Indexed Databases